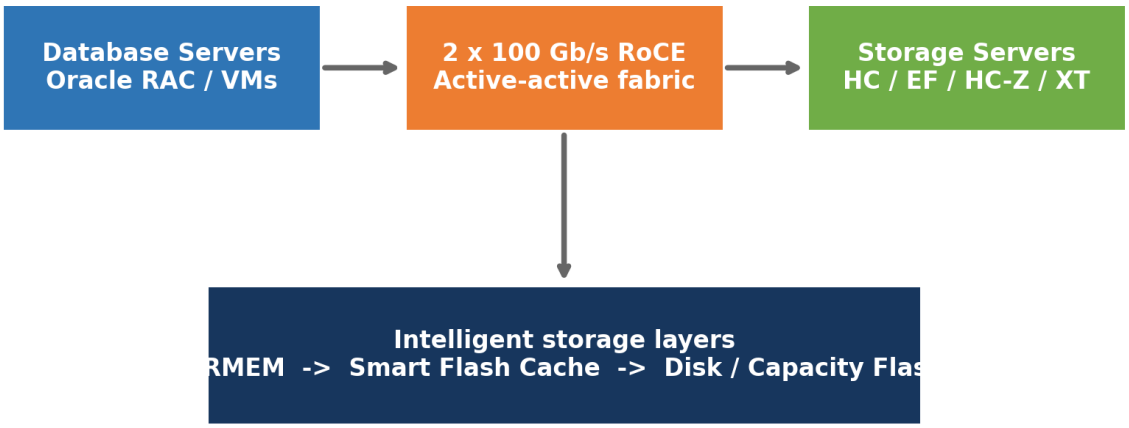
# Oracle Exadata X11M

## New Features

## Detailed Comparison with Exadata X7




X11M keeps SQL intelligence close to the data, so less data crosses the network.

# EXECUTIVE SUMMARY

Exadata X11M is a major platform change from Exadata X7. The improvement is not only a faster processor. X11M adds a much faster internal network, a shared memory cache inside storage, newer flash and disk choices, more flexible server sizes, AI-oriented SQL offload, Exascale storage, easier test and development cloning, and better control of energy use.

**Main conclusion**

An X7 customer should consider X11M when the current system is limited by CPU, I/O latency, scan speed, storage capacity, consolidation density, patching flexibility, or the need to support Oracle AI Database 26ai workloads. The final decision should be based on measured workload data and a supported sizing exercise.

- X11M can provide up to 2,880 database CPU cores per rack, *compared with up to 912 on X7.*
- X11M uses dual 100 Gb/s active-active RoCE links. *X7 uses 40 Gb/s QDR InfiniBand.*
- XRMEM adds a shared memory cache in storage, with lowest measured reads around 14 microseconds. *X7 flash latency was about 250 microseconds in the comparison used by Oracle.*
- Published single-rack X11M read performance reaches 25.2 million 8K read IOPS in a 9 database and 9 storage server configuration, *about 430% higher than the cited X7 full-rack result.*
- X11M adds Exascale, AI Smart Scan, VM Live Migration, Database Server-Z, HC-Z, and XT storage choices.

| Customer situation | Recommended direction | Reason |
|---|---|---|
| **X7 is stable and lightly used** | Keep X7 short term, but plan refresh | Upgrade value may be limited until capacity, support, or business requirements change. |
| **High CPU or RAC contention** | Prioritize X11M assessment | More cores, faster RoCE, Exafusion and RDMA path reduce processing and messaging limits. |
| **High read latency or random I/O** | Strong X11M candidate | XRMEM provides a new low-latency shared cache before flash. |
| **Large scans, reports, ETL or mixed workload** | Strong X11M candidate | 100 GB/s flash scan per HC/EF storage server and XRMEM columnar scans improve throughput. |
| **Need many test/dev copies** | Strong X11M with Exascale | Thin clones and snapshots use much less space and are easier to automate. |
| **AI Vector Search planned** | Strong X11M with 26ai | AI Smart Scan moves vector work to storage and reduces data movement. |

# 1. Scope and Important Comparison Notes

This guide focuses on bare metal and on-premises Exadata. Some software capabilities can also be used with virtualized deployments, but cloud subscription operations are outside the main scope.

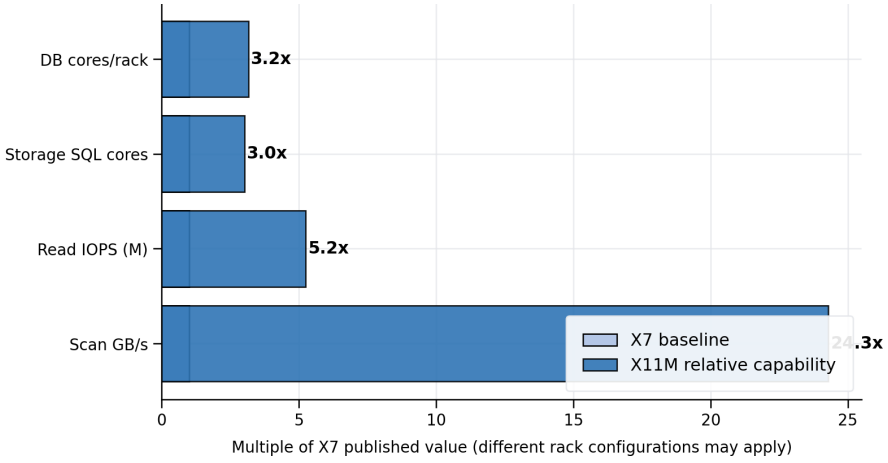
**Fair comparison note**

Oracle published X7 and X11M maximum figures using different rack layouts. Therefore, maximum-to-maximum numbers show platform capability, but they are not a guaranteed speed-up for every customer. Application SQL, database version, storage layout, compression, concurrency, and licensing all affect the result.

- X7 comparison baseline: Exadata X7-2, because this is the common two-socket X7 platform and is documented by Oracle.
- X11M values are taken from the supplied X11M data sheet, Version 1.5, dated 2026.
- Performance claims are Oracle published measurements. A proof of concept should use the customer workload.

# 2. X7 AND X11M AT A GLANCE

| Area | Exadata X7-2 | Exadata X11M | What changes for the customer |
|---|---|---|---|
| *Database CPU* | 2 x 24-core Intel Xeon per DB server; up to 912 cores/rack | **2 x 96-core AMD EPYC per DB server; up to 2,880 cores/rack** | More consolidation and parallel processing headroom. |
| *Database memory* | 384 GB standard; up to 1.5 TB/server | **512 GB to 3 TB/server; Server-Z offers 768 GB or 1,152 GB** | More room for SGA, PGA, VMs and mixed workloads. |
| *Internal fabric* | 40 Gb/s QDR InfiniBand | **2 x 100 Gb/s active-active RoCE** | Higher bandwidth and a current Ethernet-based RDMA fabric. |
| *Storage compute* | 2 x 10-core processors per X7 storage server | **2 x 32-core processors in HC/EF; 1 x 32-core in HC-Z/XT** | More storage CPU for Smart Scan and offload. |
| *HC disk per server* | 12 x 10 TB = 120 TB | **12 x 22 TB = 264 TB** | 2.2x raw disk capacity per HC server. |
| *HC flash cache* | 4 x 6.4 TB = 25.6 TB | **4 x 6.8 TB = 27.2 TB, PCIe 5.0** | Modest capacity increase, newer interface and higher throughput. |
| *EF raw capacity* | 8 x 6.4 TB = 51.2 TB | **4 x 30.72 TB = 122.88 TB capacity flash plus 27.2 TB performance flash** | About 2.4x capacity flash plus a separate large performance cache. |
| *Storage memory cache* | No XRMEM tier | **Up to 1.25 TB XRMEM per HC/EF server** | A new shared, very low-latency read tier. |
| *Read IOPS per rack* | Up to 4.8M published standard full-rack result | **Up to 25.2M published 9+9 result** | Large potential gain for cached 8K random reads. |
| *SQL scan bandwidth* | Up to 350 GB/s per full rack | **Up to 8.5 TB/s from XRMEM in a 2+17 layout** | Major analytics potential, with different test configurations. |



# 3. New Compute Platform

## 3.1 More Database CPU and Memory

Each standard X11M database server has two 96-core AMD EPYC processors, for 192 physical cores. ***X7-2 uses two 24-core Intel processors, for 48 physical cores.*** This gives **X11M four times the core count** per standard database server before considering differences in processor design and speed.

- Benefit: run more databases or VMs on fewer database servers.
- Benefit: allow more parallel query workers for data warehouse and reporting work.
- Benefit: keep OLTP, batch, reporting, and maintenance jobs on one platform with better resource separation.

**Example**

An X7 rack that is CPU-constrained during month-end processing may be able to consolidate the same workload on fewer X11M servers. However, Oracle Database licenses are tied to enabled cores, so sizing should target the cores actually needed, not simply the hardware maximum.

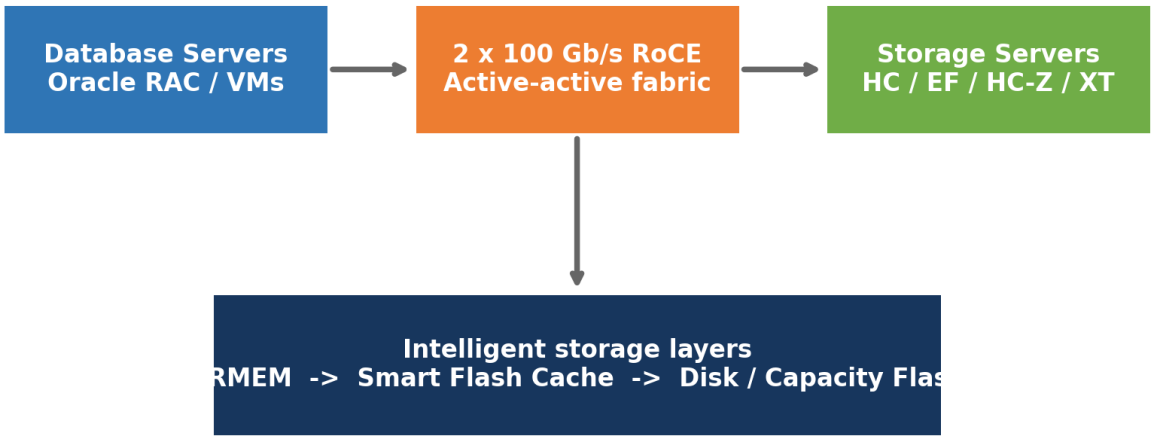
## 3.2 Database Server-Z

**X11M introduces Database Server-Z with one 32-core processor and 768 GB memory**, expandable to 1,152 GB. It is useful when a workload needs Exadata storage features but does not need 192 cores per database server.

- Good fit for small to medium databases, limited VM estates, and license-sensitive deployments.
- Can be mixed with standard X11M server choices, subject to supported configuration rules.
- Helps match CPU, memory, power, and license needs more closely.

# 4. RoCE Network and RDMA Improvements

**X7 uses a 40 Gb/s QDR InfiniBand internal network. X11M uses two active 100 Gb/s RoCE ports in each server, giving 200 Gb/s of active link capacity.** RoCE still provides Remote Direct Memory Access, so data can move between server memory areas with very little operating-system involvement.



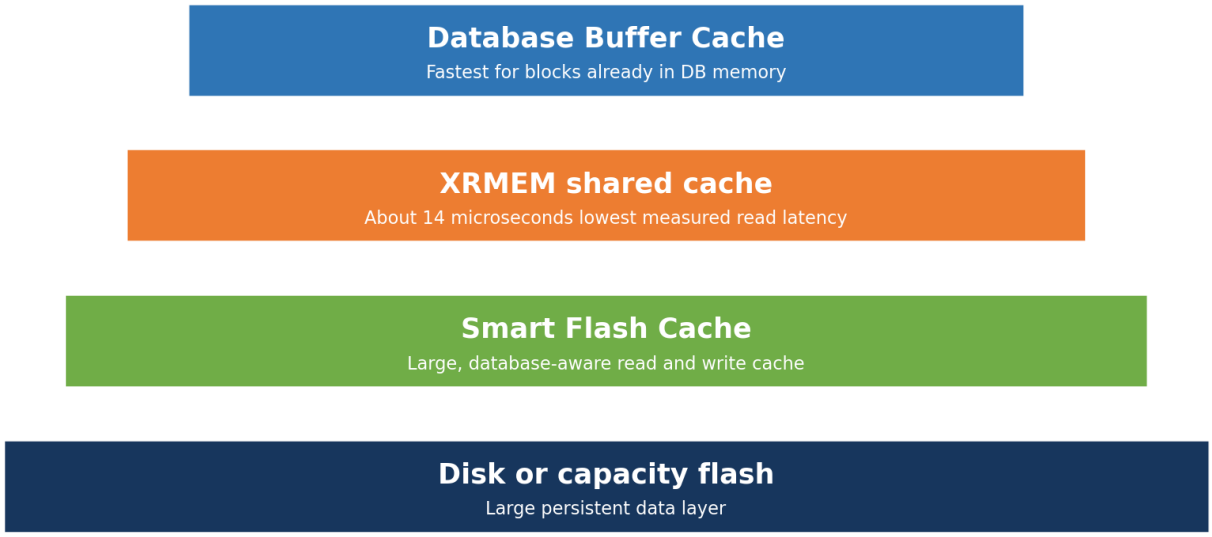

X11M keeps SQL intelligence close to the data, so less data crosses the network.

**X11M internal communication uses an active-active RDMA over Converged Ethernet fabric.**

| Feature | Simple meaning | Business effect |
|---|---|---|
| 2 x 100 Gb/s active-active RoCE | Both high-speed ports carry traffic. | More bandwidth and no idle standby link during normal operation. |
| Exafusion Direct-to-Wire | RAC messages bypass much of the normal network software path. | Lower RAC messaging overhead and better scale for busy OLTP. |
| RDMA undo reads | An RAC instance can read need undo blocks from another instance using RDMA. | Faster consistent reads for some cross-instance workloads. |
| Priority channels | Latency-sensitive traffic is separated from bulk traffic. | Backups or scans are less likely to disturb redo and Cache Fusion traffic. |
| Sub-second failure detection | The RDMA fabric helps detect failed nodes quickly. | Shorter application brownout during some server failures. |

# 5. XRMEM: The Biggest OLTP Storage Change

**Exadata RDMA Memory, called XRMEM, is memory inside the storage servers used as a shared read cache.** It sits in front of Smart Flash Cache. Database servers read this memory through RDMA, without using a normal block-I/O path.




X11M adds XRMEM between the database buffer cache and flash.

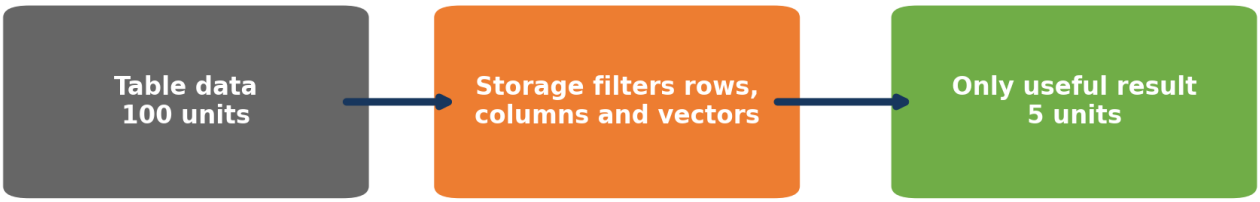
- HC and EF storage servers can provide 1.25 TB of XRMEM each. HC-Z can provide 576 GB.
- **Oracle reports lowest measured 8K read latency of about 14 microseconds from XRMEM**, compared with about 150 microseconds from X11 storage without XRMEM and **about 250 microseconds for the cited X7 flash comparison.**
- Data is automatically managed and protected across storage servers. Applications do not need code changes.
- Any database instance can benefit from the shared XRMEM tier, unlike server-local caches that may be unevenly used.

### Simple OLTP example

A payment application repeatedly reads account, customer, and reference-table blocks. **When these hot blocks are in XRMEM, the database server can obtain them much faster than reading from flash**. This can reduce wait time for storage reads and improve transaction response time.

# 6. Faster Analytics and Smart Scan

Smart Scan sends filtering and selected SQL work to storage servers. X7 already supports Smart Scan. **X11M keeps this design, but provides much more storage CPU, faster flash, a faster network, and columnar data access from XRMEM.**




Simple example: filtering in storage reduces network traffic and database-server CPU work.

**Illustrative Smart Scan flow. The values are examples, not benchmark results.**

- HC and EF storage servers can scan flash at up to 100 GB/s per storage server, 2.2 times the prior generation stated in the X11M data sheet.
- **Smart Scan of columnar data in XRMEM can reach up to 500 GB/s per HC or EF storage server.**
- A large X11M rack configuration can reach up to 8.5 TB/s XRMEM scan throughput.
- Storage Indexes avoid reading storage regions that cannot contain the requested values.

### SQL example

For a query that asks for March orders above $1,000, storage servers can read the table, remove other months and smaller orders, and return only required columns. Less data travels to database servers, and database CPUs do less work.

# 7. Storage Options and Capacity

| X11/X11M choice | Best suited for | Key capacity per server | Important note |
|---|---|---|---|
| **X11M High Capacity (HC)** | Balanced OLTP, analytics and large databases | **264 TB disk + 27.2 TB performance flash + 1.25 TB XRMEM** | Main balanced option. |
| **X11M Extreme Flash (EF)** | Very high I/O and low-latency all-flash workloads | **122.88 TB capacity flash + 27.2 TB performance flash + 1.25 TB XRMEM** | About 2.4x EF usable raw storage component versus X7 EF raw capacity. |
| **X11M High Capacity-Z** | Small to medium workloads | **132 TB disk + 13.6 TB flash + 576 GB XRMEM** | Lower CPU and capacity entry point. |
| **X11M Extended (XT)** | Archive, retention and less-used online data | **264 TB disk + 13.6 TB flash** | Can use Exascale pool or separate ASM disk group; advanced smart features need an optional license. |
| **X11 variants without M** | Cost-sensitive growth | **Same main disk/flash layout, without XRMEM by default** | Memory can be added later to enable XRMEM. |

## 7.1 Why XT Matters

**X7 offered HC and EF choices. X11M adds a purpose-built XT tier for older or less frequently accessed data that still must remain online.** This helps avoid buying premium storage for every byte.

- Use for historical partitions, compliance data, images, documents, JSON, XML, LOBs, and backups where suitable.
- **Use Automatic Data Optimization and Hybrid Columnar Compression to move aging data to a lower-cost tier.**
- **Keep one Exadata operating model instead of introducing a separate storage platform.**

# 8. EXADATA EXASCALE

Exascale is a newer storage architecture that separates database compute from storage management more cleanly. Databases use storage vaults that are automatically spread across available storage. It is designed for flexible growth, shared use, and cloud-style operations on Exadata.

| Capability | How it works | Customer value |
|---|---|---|
| **Thin clones** | Clones share unchanged blocks with the source and store only changed data. | Many development copies require much less storage. |
| **Read/write source support** | A clone can be made from an active read/write database or PDB. | No need to make the production source read-only for cloning. |
| **Independent clones** | The source can continue, change, or even be removed without breaking an independent clone. | More flexible test and development life cycles. |
| **PDB Snapshot Carousel** | Snapshots can be created automatically at regular intervals. | Easy point-in-time test copies. |
| **REST and gDBClone automation** | Cloning can be integrated with CI/CD and command-line workflows. | Faster application testing and release cycles. |
| **Shared VM boot volumes** | Exascale volumes can be reached from database server hosts. | Supports VM mobility and live migration. |

**Example**

A 20 TB production database is needed by six test teams. Traditional full copies could require about 120 TB before growth. Thin clones share common blocks, so storage use is mainly the source plus each clone's changed blocks. Actual savings depend on how much each clone changes.

# 9. AI SMART SCAN FOR ORACLE AI DATABASE 26AI

X11M is designed for Oracle AI Database 26ai vector workloads. AI Smart Scan performs vector-distance calculations and Adaptive Top-K filtering in storage, near the vector data. This reduces the number of candidate rows sent to database servers.

- Oracle states AI Smart Scan can accelerate AI Vector Search by up to 30 times for supported cases.
- The data sheet also states up to 4.6 times faster queries and up to 4.7 times more data filtering from the storage-side optimizations.
- BINARY vector dimension formats can provide up to 32 times faster AI queries in the stated comparison.

**Simple example**

A support portal searches one million document embeddings for the ten most similar documents. Storage servers calculate distances in parallel and return the strongest candidates instead of sending all vectors to database servers.

# 10. AVAILABILITY, MAINTENANCE, AND SECURITY

| Area | X11M capability | Why it matters for an X7 customer |
|---|---|---|
| **VM Live Migration** | **Moves active RDMA-enabled database VMs between database servers with little or no application disruption.** | Planned maintenance and workload balancing can be easier. |
| **Instant failure detection** | **RDMA-based sub-second detection for some node failures.** | Reduces the time applications wait for failure recognition. |
| **I/O latency capping** | **Redirects unusually slow reads or writes to healthy copies or devices.** | Reduces long I/O outliers. |
| **Predictive failure handling** | **Uses learned patterns and health data to remove risky components gracefully.** | Lowers risk of an unexpected component issue. |
| **Exadata Live Update** | **Applies supported system and security fixes while the OS remains online.** | Can reduce maintenance disruption. |
| **Secure Boot and Secure Erase** | **Checks trusted boot components and cryptographically erased storage devices.** | Helps protect the platform and data during reuse or retirement. |
| **Full-stack integration** | **Oracle tests firmware, OS, storage software and database together.** | Simplifies support and reduces compatibility risk. |

# 11. Energy and Data Center Efficiency

X11M provides more control over power use than X7-era hardware. Customers can select the 32-core Server-Z or the 192-core standard server, limit active database cores with Capacity-on-Demand, set CPU power targets, and allow power to change with workload demand.

- Choose fewer active cores when the workload does not need the full server.
- Consolidate older racks to reduce rack space, network devices, cooling, and administration.
- Use lower-power periods at night or weekends where supported operating policies allow it.

**Important**

Energy savings depend on the final X11M configuration and utilization. Compare measured X7 rack power and cooling with the proposed X11M bill of materials, not only the maximum values in a data sheet.

# 12. WHY A CUSTOMER SHOULD UPGRADE

| Upgrade reason | What X11M provides | Expected business result |
|---|---|---|
| **Faster customer transactions** | XRMEM, RoCE, more CPU and improved redo/RAC paths | Lower response time and more transactions during peaks. |
| **Shorter reports and batch windows** | More storage CPU, faster flash scan, XRMEM columnar scan | Faster close, ETL, billing, reporting and analytics. |
| **Database growth** | Larger disks, larger EF drives, up to 17 storage servers per rack | More capacity without adding a separate platform. |
| **Consolidation** | Up to 3 TB memory and 192 cores per standard DB server | Fewer systems and simpler operations. |
| **Modern development** | Exascale thin clones, snapshot workflows and APIs | Faster test environment delivery and lower copy storage. |
| **AI adoption** | AI Smart Scan and Oracle AI Database 26ai support | Run relational, JSON, vector and other data together. |
| **Lifecycle risk** | Current-generation hardware, security and management features | Reduced aging-hardware and support risk. |
| **Sustainability** | Power targets, server choices and consolidation density | Potential reduction in power, space and cooling. |

# 13. When Not to Upgrade Immediately

- The X7 system has enough capacity and meets all service levels.
- A required application or database release is not yet certified for the planned X11M software stack.
- License cost for the proposed enabled cores is not justified by the workload.
- The business cannot provide a safe migration window, and no online migration method meets the requirement.
- A refresh is planned, but network, power, cooling, rack, or data-center readiness is incomplete.

**Practical approach**

If an immediate upgrade is not justified, create a 12-to-24-month refresh plan. Track CPU, DB time, read latency, flash-cache hit rate, scan throughput, storage growth, component incidents, patch support, and license demand.

# 14. Suggested Upgrade Assessment




Use Real Application Testing, RMAN, Data Guard or logical migration methods based on downtime needs.

**Recommended high-level upgrade path from X7 to X11M.**

| Stage | Key activities | Main output |
|---|---|---|
| **1. Assess** | Collect AWR, ASH, ExaWatcher, capacity, incidents, database versions, VM layout and license usage. | Current-state baseline and pain points. |
| **2. Size** | Choose Server or Server-Z; HC, EF, HC-Z or XT; XRMEM; memory; enabled cores; network adapters. | Supported bill of materials and cost model. |
| **3. Test** | Replay workload, test SQL plans, validate throughput, failure behavior, backup and monitoring. | Proof-of-concept results and acceptance criteria. |
| **4. Migrate** | Select RMAN restore/duplicate, Data Guard switchover, transportable methods, GoldenGate, or another supported approach. | Detailed runbook and rollback plan. |
| **5. Validate** | Compare response time, DB time, waits, IOPS, scan rate, redo rate, batch time, and business transactions. | Signed performance and functional validation. |
| **6. Optimize** | Tune resource plans, Smart Flash Cache policies, consolidation, Exascale cloning and monitoring. | Stable production operation and X7 retirement plan. |

# 15. Workload-Based Decision Guide

| Workload | X7 common limit | X11M feature to test | Success measure |
|---|---|---|---|
| **High-volume OLTP** | **Flash read latency, CPU, RAC messaging** | XRMEM, RoCE, more CPU, Exafusion | P95/P99 transaction time, log file sync, gc waits, IOPS. |
| **Data warehouse** | **Scan time and database CPU** | 100 GB/s storage scan, XRMEM columnar scan, more storage CPU | Query elapsed time, bytes offloaded, scan throughput. |
| **Mixed consolidation** | **Resource interference** | IORM, Database Resource Manager, priority channels, more memory | SLA by database, CPU and I/O fairness. |
| **Test and development** | **Copy time and storage cost** | Exascale thin clones and snapshot automation | Clone time, storage consumed, environment delivery time. |
| **Vector search** | **Large vector scans and data movement** | AI Smart Scan, Adaptive Top-K, XRMEM | Search latency, recall, concurrency, DB CPU. |
| **Archive and retention** | **Expensive online capacity** | XT storage, HCC, ADO | Cost per usable TB and retrieval SLA. |

# 16. Final Recommendation

For most production X7 estates, X11M should be treated as a full platform refresh, not a simple like-for-like hardware swap. The strongest value comes when the upgrade also improves consolidation, storage tiering, test and development cloning, AI readiness, lifecycle support, and energy use.

**Recommended decision rule**

Move forward when measured X7 constraints or lifecycle risk can be clearly linked to X11M capabilities, and when a proof of concept confirms the expected result. Avoid buying the maximum configuration by default. Select the smallest supported X11M shape that meets performance, availability, growth, and license needs.

In Oracle Exadata storage terminology:

## HC = High Capacity

HC is designed to provide a **large amount of storage while still giving very good database performance**.

- Uses **traditional SAS hard disks** for large-capacity storage.
- Uses **NVMe Flash** as Smart Flash Cache.
- In X11M HC, **XRMEM** adds an even faster memory-cache layer.
- Best suited for **mixed OLTP + Data Warehouse workloads**, large databases, and customers who need both capacity and performance.
- One X11M HC Storage Server has **12 × 22 TB disks = 264 TB raw disk**, **27.2 TB performance flash**, and up to **1.25 TB XRMEM**.

Think of HC as:

**Large Disk Capacity + Flash Cache + XRMEM**

```
Database
↓
XRMEM ← Very fast memory cache
↓
Smart Flash ← Fast NVMe cache
↓
SAS Disks ← Large capacity
```

## EF = Extreme Flash

EF is designed for customers who need **maximum storage performance and very low latency**.

- It is an **all-flash storage configuration**.
- There are no SAS data disks like HC.
- X11M EF combines high-capacity NVMe flash with separate performance-optimized NVMe flash.
- XRMEM sits in front of flash for extremely fast reads.
- Best suited for **high-volume OLTP, very I/O-intensive databases, and workloads where storage latency is critical**.

- One X11M EF Storage Server provides **122.88 TB capacity-optimized flash**, **27.2 TB performance-optimized flash**, and up to **1.25 TB XRMEM**.

Think of EF as:

**All Flash + Performance Flash Cache + XRMEM**

```
Database
↓
XRMEM
↓
Performance Flash
↓
Capacity Flash
```

**HC vs EF in one line**

| Type | Full Form | Main Goal | Data Storage |
|---|---|---|---|
| **HC** | High Capacity | Balance **capacity + performance + cost** | Disk + Flash + XRMEM |
| **EF** | Extreme Flash | **Maximum I/O performance and low latency** | All Flash + XRMEM |